

Wetting-angle-kit: a Python package for automated wetting contact angle analysis of nanodroplets

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Summary

Wetting-angle-kit is a Python toolkit designed to extract wettability properties, specifically the contact angle of a droplet on a surface, from molecular dynamics (MD) simulations of interfaces between liquids and solid surfaces.

It supports a variety of standard file formats including extended XYZ, LAMMPS, and ASE-readable trajectories and offers two distinct computational methods for contact angle analysis. Furthermore, the package includes robust utilities for statistical post-processing and data visualization, providing a comprehensive workflow for wettability studies. This integrated approach reduces the need for custom analysis scripts and improves reproducibility across different simulation setups.

Statement of need

Building upon foundational work (Hautman et al., 1997), the methodologies for computing contact angles from MD simulations have progressed through several key milestones (Carlson et al., 2024; Rafiee et al., 2012; Vega & Coninck, 2016). Despite these advancements, the field currently lacks a standardized, unified tool for comparing and validating the diverse methods used to derive contact angles. Such fragmentation undermines collaborative research and reproducibility, as many implementations remain inaccessible or poorly documented. In addition, the lack of a standardized framework makes it difficult to benchmark different approaches or assess the impact of methodological choices.

Wetting-angle-kit addresses this critical gap by providing a flexible, open-source package. It enables the implementation of novel post-processing algorithms for the extraction and calculation of contact angle, to compare them against accepted techniques, and to establish a standardized benchmark for MD wettability analysis.

State of the field

General-purpose MD simulation post-processing tools, such as OVITO (Stukowski, 2010), MDSuite (Tovey et al., 2023), and MDAnalysis (Gowers et al., 2016; Michaud-Agrawal et al., 2011), provide flexible frameworks for analyzing and visualizing trajectories. However, they do

38 not include a standardized implementation of contact angle extraction methods, which are
39 typically developed as custom scripts tailored to specific systems.

40 Existing approaches to contact angle computation range from geometric fitting techniques
41 based on spherical or cylindrical cap approximations (Hautman et al., 1997) to density-based
42 interface analysis (Vega & Coninck, 2016) and pressure-tensor approaches derived from planar
43 equilibrium simulations (Carlson et al., 2024), making direct comparison across methodologies
44 challenging. Wetting-angle-kit complements existing tools by focusing specifically on wettability
45 analysis and providing a consistent environment for comparing multiple methods. This design
46 promotes reproducibility and facilitates the development and/or implementation of other
47 methods.

48 Software design

49 Wetting-angle-kit is organized into three main components: parsers, analysis, and visualization,
50 as represented in Figure 1. This modular organization separates data handling, analysis, and
51 visualization, allowing components to evolve independently while simplifying the integration of
52 new features.

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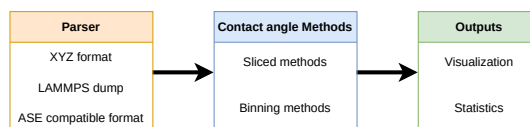


Figure 1: Wetting-angle-kit package structure. The package is composed of three main modules: parsers, analysis, and visualization.

53 The parser module provides a unified interface for reading trajectory data from multiple formats,
 54 ensuring consistent handling of atomic coordinates, simulation boxes, and frame information.
 55 This abstraction ensures that analyses are independent of the input format, enabling consistent
 56 workflows across different simulation engines. The parser leverages established trajectory-
 57 reading tools when available, while extended XYZ parsing is implemented directly within the
 58 package. The parser also consistently handles periodic boundary conditions, ensuring that
 59 droplet shapes are correctly reconstructed across simulation boundaries and avoiding artifacts
 60 in interface detection. This consistency facilitates seamless integration with downstream
 61 analysis methods, enabling researchers to easily incorporate support for additional file formats
 62 or simulation engines.

63 The analysis module implements two complementary approaches for contact angle computation
 64 that are illustrated in Figure 2. The slicing method consists in a frame-by-frame geometric

analysis, which enables a detailed temporal resolution. In practice, this approach provides a local characterization of the liquid-vapor interface, allowing the detection of asymmetries and transient deformations of the droplet shape. It is particularly well suited for non-equilibrium simulations or systems where the droplet deviates from an ideal spherical cap. In contrast, the binning method constructs time-averaged density fields, reducing thermal fluctuations and producing a smoother and more stable interface. This makes this approach suitable for extracting equilibrium contact angles from noisy datasets. However, this temporal averaging may obscure short-lived fluctuations and local deviations from ideal geometries. The binning method is also more suited to symmetric systems, since atoms are folded into a single quadrant. Due to the finer analysis it provides, the slicing method is one order of magnitude more expensive computationally than its binning counterpart. These two approaches reflect a trade-off between temporal resolution and statistical robustness, allowing users to select the method best suited to their system.

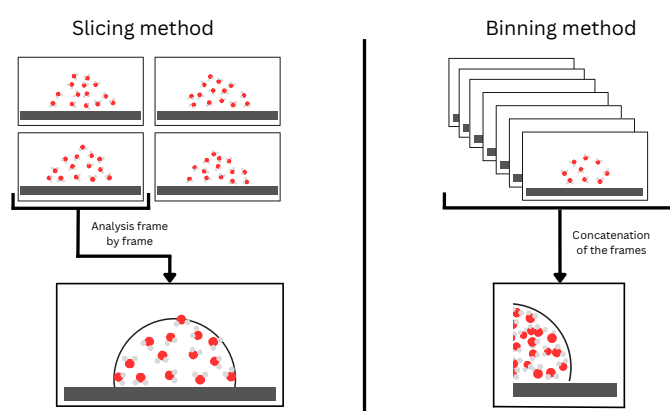


Figure 2: Schematic representation of the two methods developed in wetting-angle-kit to compute contact angle from a MD trajectory. In the slicing method (left), all trajectory frames are analyzed and a circle is fitted on each of those, providing a time evolution of the contact angle. In the binning method (right), all frames are concatenated to fictitiously increase the molecular density of the droplet, allowing for smoother statistics at the cost of losing the time dependence of the contact angle.

Additionally, wetting-angle-kit supports two geometric models commonly used in the literature for droplets: spherical and cylindrical (Scocchi et al., 2011) (see Figure 3). While the spherical case provides a more direct representation of the droplet curvature, a cylindrical geometry reduces curvature effects and computational cost by relying on periodic boundary conditions along the cylinder axis, at the expense of relying on an idealized geometry.

Geometric representations of droplets used in the analysis: spherical droplet (left) and cylindrical droplet (right).

Figure 3: Geometric representations of droplets used in the analysis: spherical droplet (left) and cylindrical droplet (right).

The visualization module includes tools to support interpretation and validation of analysis

84 results without requiring external post-processing tools. These tools consists in (1) a contact-
85 angle vs. trajectory timeframe for the slicing analysis, and (2) a density heatmap based on the
86 binning analysis.

87 The software architecture relies on abstract base classes to enforce consistent interfaces and
88 facilitate extensibility. This design enables users to implement new computational methods while
89 maintaining compatibility with existing workflows, promoting reuse and method comparison. It
90 also facilitates the integration of newly developed methods into an existing and standardized
91 analysis pipeline.

92 Research impact statement

93 Wetting-angle-kit provides a reproducible framework for contact angle analysis in MD
94 simulations, addressing a common need in studies of nanoscale wetting. The package has
95 been validated using MD simulations of water droplets on graphene and polymer substrates,
96 yielding contact angle values consistent with literature results (e.g., $\sim 93^\circ$ for graphene, $\sim 110^\circ$
97 for PTFE), see Figure 4. The reported contact angles were obtained by analyzing droplets
98 of increasing sizes and extrapolating to the macroscopic limit using the modified Young's
99 equation ref, where the contact angle is related to droplet size, enabling the estimation of
100 the infinite-droplet contact angle through linear extrapolation. These results are consistent
101 with values reported in the literature, obtained using similar interatomic potential parameters
102 (Jorgensen et al., 1996) for the MD simulation.

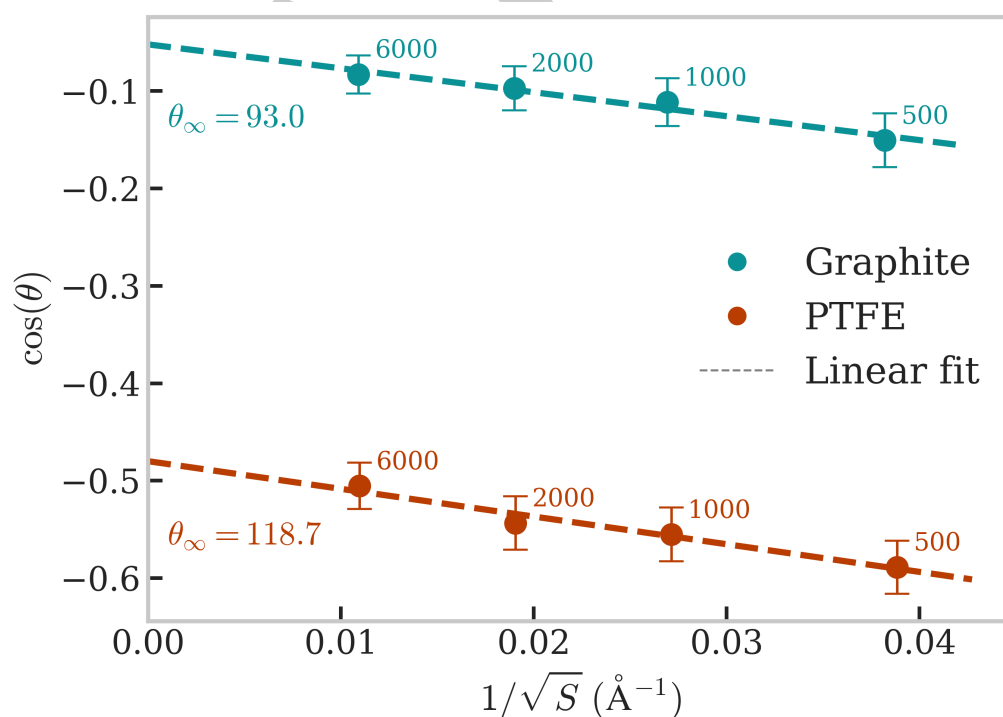


Figure 4: Size-dependent contact angle analysis for water droplets on graphite and PTFE substrates. Values of $\cos(\theta)$ are plotted as a function of the inverse square root of the droplet surface area for droplets containing between 500 and 6000 water molecules. A linear extrapolation following the modified Young's equation is used to estimate the macroscopic (infinite-droplet) contact angle.

103 By enabling systematic comparison of analysis methods and providing standardized workflows,
104 wetting-angle-kit supports more robust and reproducible wettability studies. Its modular design
105 also facilitates integration into existing simulation pipelines and encourages community-driven

106 extensions. The package is expected to be particularly useful for researchers using various types
107 of force fields (classical, ab initio, and machine learned) or investigating nanoscale interfacial
108 phenomena.

109 AI usage disclosure

110 Generative AI tools (Claude Code with Sonnet 4.6 and Opus 4.7, **XXX Gabriel add yours**)
111 were used in the development of the software, for drafting and assisting debugging. Generative
112 AI was used to assist in refining the language, translation and clarity of the manuscript and
113 docstring. All AI-assisted contributions were verified and approved by the authors.

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